Challenge 9: Union-Closed Sets Conjecture

Definition 1 (Union-closed set)**.** A set $\mathcal{F}$ of sets is *union-closed* if for any two sets in $\mathcal{F}$, their union is also in $\mathcal{F}$. For a nonempty finite set $\mathcal{F}$, the *density* of an element x with respect to $\mathcal{F}$ is defined to be

$$\frac{|\{F \in \mathcal{F} : x \in F\}|}{|\mathcal{F}|}$$

Conjecture 1.1 (Union-closed sets conjecture [1])**.** For every finite union-closed set $\mathcal{F} \notin \{\emptyset, \{\emptyset\}\}$, there exists an element x of density at least one half.

We are interested in the following parametrized version of the conjecture.

Conjecture 1.2 (Scaled union-closed sets conjecture)**.** Let $r \in \mathbb{N}$. For every finite union-closed set $\mathcal{F} \notin \{\emptyset, \{\emptyset\}\}$, there exists an element x of density at least $\frac{1}{2} - \frac{1}{r+2}$.

For $r = 0$ the bound equals 0 and the statement is elementary, while as $r \to \infty$ the bound approaches $\frac{1}{2}$ and the statement recovers the full conjecture in the limit. The best published lower bound on the density (≈ 0.382, obtained via the entropy method) settles the cases $r \leq 6$; the conjecture is open for all $r \geq 7$.

References

[1] N. Nagel. *Notes on the Union Closed Sets Conjecture*. 2023. arXiv: `2208.03803 [math.CO]`. URL: `https://arxiv.org/abs/2208.03803`.